Protecting $11.5M of gross margin from supplier disruption

NorthBay Resilience: supply network stress testing for NorthBay Foods

For: Dana (VP Supply Chain) · Procurement leadership · Supply Chain Analytics

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Executive summary

- **$11.5M of gross margin (15% of the year) and $39.8M of revenue** are exposed to single-supplier disruptions.
- **71% of the exposure sits with three suppliers**: Midwest Sweeteners, Kerala Spice Traders, FlavorCraft Blends.
- **Two moves cut margin at risk by 68%, to $3.7M:**
  1. Qualify GulfCane Syrups as a second sweetener source for cereal and granola bars.
  2. Hold a 6-week seasoning buffer for Kettle Chips.
- **≈ $0.8M/yr expected margin protected for ≈ $150k one-off + $36k/yr, so payback is under 3 months.**
- Planners now answer multi-tier "what-ifs" in **~30 seconds instead of 1–2 weeks**.

Ask: a 6-week pilot on NorthBay's real ERP / SRM data.

The problem: disruptions are a surprise, and so is their cost

- Illustrative history: a refinery explosion, a seasoning-line fire and a Gulf hurricane each forced emergency re-planning.
- Nobody could answer **before** the event:
  - Which suppliers can we not live without, and for how long?
  - How much margin does each disruption cost after our best response?
  - Which mitigation is worth paying for: second source, buffer stock, or nothing?
- Today's answer is a 1–2 week spreadsheet exercise across ERP, SRM, WMS and planning data, so it arrives after the money is gone.

Where NorthBay is exposed

Each supplier is down for its typical recovery time, and the network is re-planned optimally.

#	Supplier	Typical disruption	Down	We cope for	Gross margin lost	Products hit
1	Midwest Sweeteners	Refinery explosion	9 wk	3.6 wk	**$2.99M**	Morning Oats Cereal, Trail Granola Bar
2	Kerala Spice Traders	Monsoon flooding	12 wk	5.3 wk	**$2.76M**	Kettle Chips
3	FlavorCraft Blends	Blending-line fire	10 wk	3.8 wk	**$2.42M**	Kettle Chips
4	BarrierFilm Co.	Labor strike	7 wk	4.3 wk	$0.68M	Trail Granola Bar
5	Citrus Essence Labs	Citrus crop loss	7 wk	5.6 wk	$0.55M	Kettle Chips

11 of 24 suppliers cause unavoidable loss. **All 5 plants** have at least one single-sourced input (11 in total).

Multi-supplier events the risk team worries about

Event	Suppliers down	Duration	Gross margin lost	Products hit
Wisconsin film converters shut down	FlexiWrap, BarrierFilm	5 wk	**$1.93M**	Chips, Cereal, Granola Bar
Midwest drought hits corn sweetener	Iowa & Illinois corn	9 wk	**$1.69M**	Morning Oats Cereal
North American PET resin shortage	Gulf Coast, Monterrey resin	6 wk	$0.61M	Fizz Cola
Gulf Coast hurricane	GulfCane, Gulf Coast Resins, Bayou Sugar	8 wk	$0.56M	SunValley Orange Drink

Any combination can be asked in the app in plain English, e.g. "*what if both film converters go down for 5 weeks?*"

What it is worth to fix

The full portfolio is re-run for each option.

Option	Margin at risk	Reduction	Cost
Today	$11.49M	–	–
A. Second sweetener source (GulfCane → cereal, bars)	$8.86M	23%	≈ $150k qualification
B. 6-week seasoning buffer at FlavorCraft	$6.33M	45%	≈ $144k inventory, $36k/yr carrying
C. Second seasoning source (SpiceWorks → chips)	$11.36M	**1%**	≈ $150k: *not worth it*
A + B: recommended	**$3.70M**	**68%**	**≈ $150k one-off + $36k/yr**

The intuitive option C fails because SpiceWorks has no spare capacity. The optimizer sees it; a spreadsheet does not.

Value for each stakeholder

Stakeholder	What changes	KPI
VP Supply Chain (sponsor)	Margin at risk becomes a managed number reviewed monthly	$11.5M → **$3.7M** margin at risk; largest single exposure $2.99M → $0.68M
Procurement (owner)	A ranked, costed list of dual-sourcing and buffer decisions	Single-sourced inputs 11 → 9; next wave: packaging film & cartons
Supply planners	Answer "what if X is down N weeks?" in the S&OP meeting	1–2 weeks → **~30 seconds**
Finance	Expected-loss view for risk reserves and insurance	≈ **$0.8M/yr** expected margin protected, payback < 3 months

How it works

1. **Ingest** ERP, SRM, PLM, WMS and APS data continuously (Lakeflow), with automatic data-quality checks.
2. **Govern** one trusted network model with lineage and access control (Unity Catalog).
3. **Stress-test** every supplier with an optimization model that re-plans the whole network (Databricks jobs).
4. **Serve** results and the live network to the app in milliseconds (Lakebase).
5. **Ask** questions in plain English: Genie answers data questions, and the AI agent runs what-ifs on the optimizer.
6. **Act** in one business app: risk overview, AI assistant, and an audit log of every scenario.

Why you can trust the answers

- **Numbers come from math, not the AI.** The optimizer computes every impact figure, and the AI explains it and proposes actions.
- **Same question, same answer.** The optimization is deterministic and every scenario is logged.
- **Clean data in.** Duplicate supplier records, negative inventory scans and links to unknown suppliers are caught and counted.
- **Governed end to end.** Lineage runs from source files to the app. The app sees only the curated gold data.
- **Evaluated.** 8-question test set: correct tool choice 8/8, safety 8/8, actionable advice 8/8, and off-topic questions declined.

Pilot plan: 6 weeks

Week	Milestone
1–2	Connect SAP (plants, BOM, inventory) and the supplier portal with Lakeflow Connect; validate the network model with planners
3	Calibrate recovery times with procurement; run the portfolio stress test on real data
4	Procurement reviews the top-10 exposures and costs the mitigation options
5–6	Planners use the app in two S&OP cycles; success review

Success criteria: the top-3 exposures match planners' intuition or surprise them with a defensible reason; at least one mitigation decision is taken on the numbers; the what-if answer is under 1 minute.

Appendix: assumptions

- Data is synthetic for NorthBay Foods (fictional); structure and economics are realistic for a mid-size US CPG company.
- Gross margin at risk = the sum over single-supplier disruptions at each supplier's typical recovery time (not simultaneous).
- Expected value: a 10% annual probability of each supplier's typical disruption (≈ 2–3 material events per year across 24 suppliers).
- Costs: supplier qualification ≈ $150k; seasoning at $0.40 per case-equivalent; 25% inventory carrying cost.
- Planner time: ~25 what-if analyses per year × 40 hours saved.